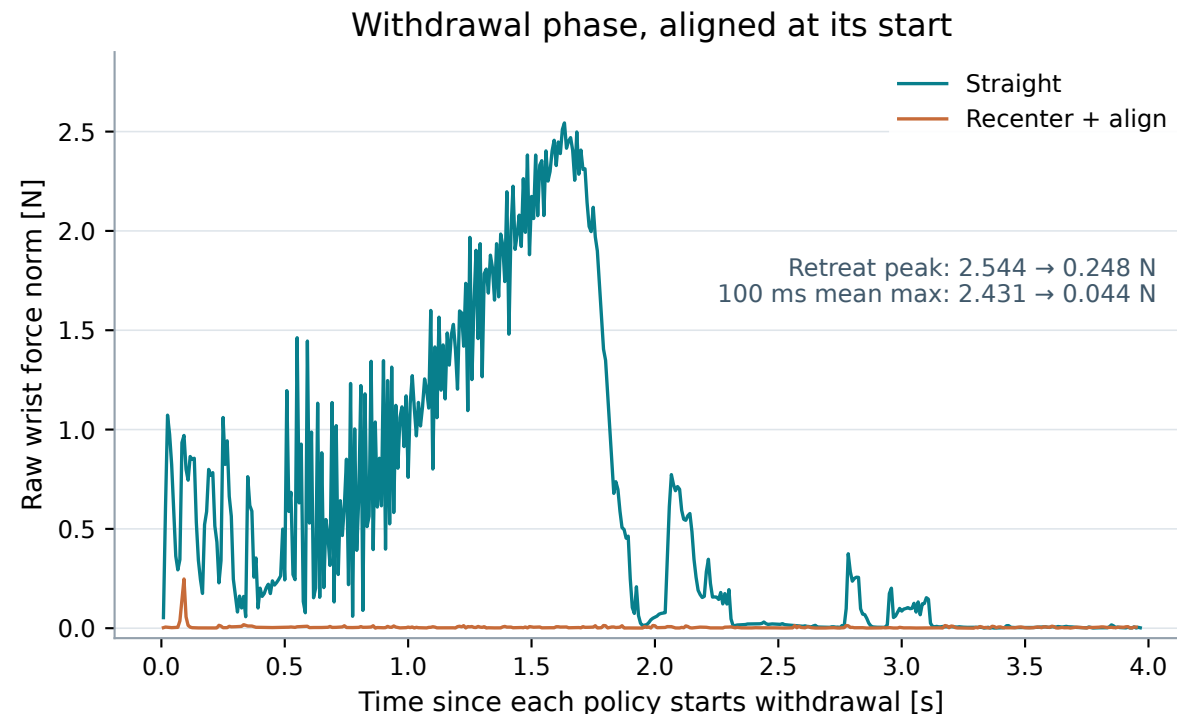
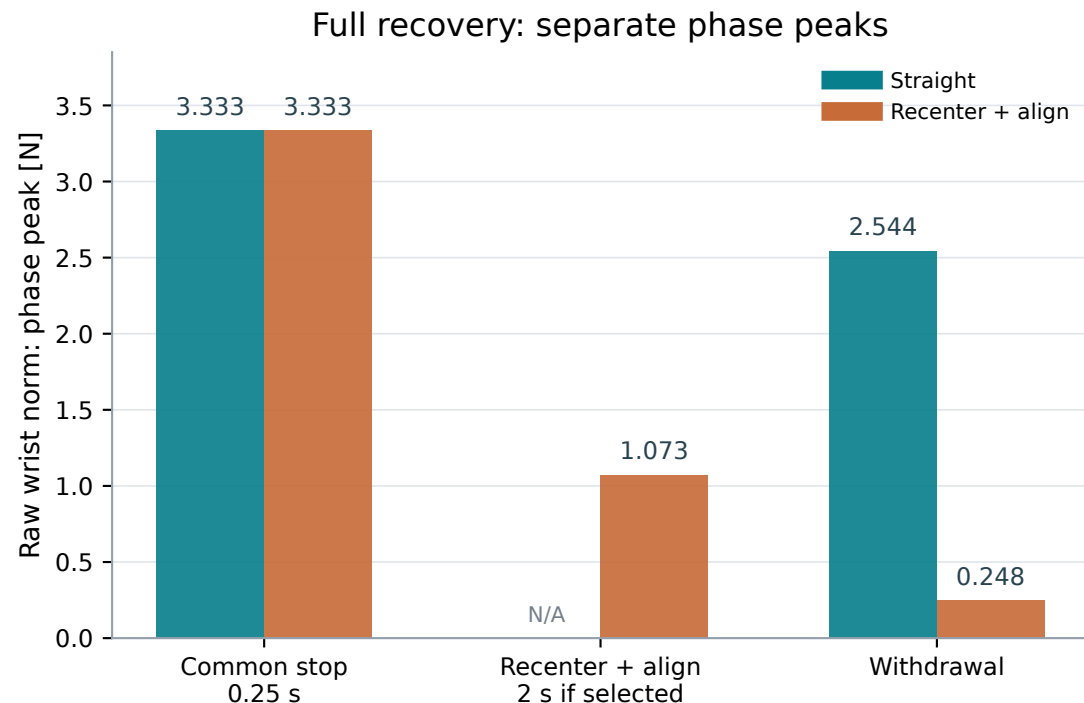


Recenter + align reduces the subsequent withdrawal load

0.2 mm radial clearance · 2° insertion tilt · same terminal checkpoint · both replays matched and recovered safely

The pose adjustment moves the peg toward the hole center in XY and returns its orientation upright.



Whole-recovery peak remains 3.333 N for BOTH policies, caused by the common stop stage.

Total recovery duration: straight 4.217 s; recenter + align 6.200 s. The stop holds the attained hand pose and can unload insertion forces.

The reduction belongs to the combined centering/orientation policy; this comparison does not isolate the effect of angle alone.

These are independent replay branches with observable-state matching. Numerical screening passed; hidden contact memory and timestep convergence are not certified.